

Topic 8. Interpolation



8.1 Introduction.

Models are constructed based on experimental data.

This data is obtained in the form of tables $\{X_i, Y_i\}$.

However, in order to use tabular data in equations, they must be represented as functions.

For this purpose, tabular data is approximated by the function (polynomial)

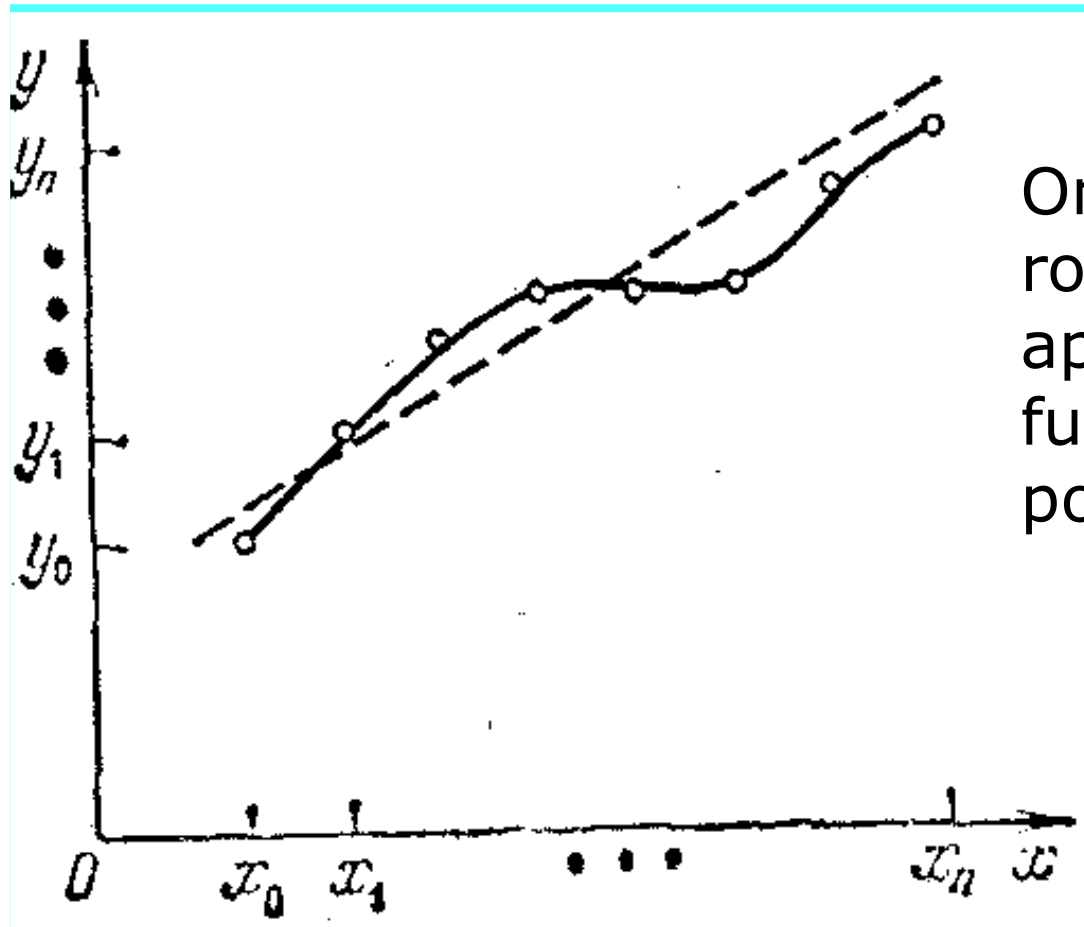
$$\phi(x) = a_0 + a_1x + a_2x^2 + \dots + a_mx^m \quad (8.1)$$

The coefficients of the polynomial are calculated from the equations

$$\varphi(x_i) = y_i, \quad i = 0, 1, \dots, n \quad (8.2)$$

So, when interpolating, the main condition is to pass the graph of the interpolation polynomial through the values of the functions at the interpolation nodes.

Sometimes interpolation is difficult or impossible. A way out of this situation can be found by choosing a polynomial whose graph passes close to these points.



One of these types is the root-mean-square approximation of functions using the polynomial (8.1).

Interpolation and approximation

$$f(x)$$

8.2 Linear and quadratic interpolation

The simplest and most frequently used type of local interpolation is linear interpolation.

It consists in the fact that given points (X_i, Y_i) ($i=0,1,\dots,n$) are connected by straight lines

Since there are n intervals (x_{i-1}, x_i) ,
then for each of them, the equation of a straight
line passing through two points is used as the
equation of the interpolation polynomial.

In particular, for the i -th interval we can write the
equation of a straight line passing through the points
 (x_{i-1}, y_{i-1}) **And** (x_i, y_i) , as

$$\frac{y - y_{i-1}}{y_i - y_{i-1}} = \frac{x - x_{i-1}}{x_i - x_{i-1}}$$

and we find

$$y = y_{i-1} + \frac{(x - x_{i-1})(y_i - y_{i-1})}{x_i - x_{i-1}}$$

Therefore, when using linear interpolation, you first need to determine the interval in which the value of the argument x falls, and then find the approximate value of the function at that point.

Let us now consider the case of quadratic interpolation.

As an interpolation function on a segment $[x_{i-1}, x_{i+1}]$ a quadratic trinomial is accepted.

This interpolation is also called parabolic.

Quadratic Trinomial Equation

$$y = a_i x^2 + b_i x + c_i \quad x_{i-1} \leq x \leq x_{i+1} \quad (8.3)$$

contains three unknown coefficients

a_i, b_i, c_i which require three equations to determine.

They are the conditions for the passage of a parabola through three points

$$(x_{i-1}, y_{i-1}), (x_i, y_i), (x_{i+1}, y_{i+1}).$$

These conditions can be written in the form

$$\begin{cases} a_i x_{i-1}^2 + b_i x_{i-1} + c_i = y_{i-1} \\ a_i x_i^2 + b_i x_i + c_i = y_i \\ a_i x_{i+1}^2 + b_i x_{i+1} + c_i = y_{i+1} \end{cases} \quad (8.4)$$

8.3 Lagrange polynomial.

Let's consider the case of global interpolation, i.e. construction of an interpolation polynomial, common for the entire segment $[x_0, x_n]$.

In this case, the graph of the interpolation polynomial must pass through all given points.

There are simpler algorithms for constructing interpolation polynomials.

We will look for a polynomial in the form of a linear combination of polynomials of degree n

$$L(x) = y_0 l_0(x) + y_1 l_1(x) + \dots + y_n l_n(x) \quad (8.5)$$

In this case, we require that each polynomial $l_i(x)$

vanished at all nodes
interpolation, with the exception of one (i-th), where
it should be equal to one.

It is easy to check that these conditions are satisfied by a polynomial of the form

$$l_0(x) = \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)}. \quad (8.6)$$

Really, $l_0(x_0) = 1$ at $x = x_0$.

At $x = x_1, x_2, \dots, x_n$

the numerator of the expression goes to zero.

By analogy with (8.6) we obtain

$$l_1(x) = \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)}$$

$$l_2(x) = \frac{(x - x_0)(x - x_1)(x - x_3) \dots (x - x_n)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3) \dots (x_2 - x_n)}$$

.....

$$l_i(x) = \frac{(x - x_0) \dots (x - x_{i-1})(x - x_{i+1}) \dots (x - x_n)}{(x_i - x_0) \dots (x_i - x_{i-1})(x_i - x_{i+1}) \dots (x_i - x_n)} \quad (8.7)$$

.....

Substituting the resulting expressions into (8.5), we find

$$L(x) = \sum_{i=0}^n y_i \frac{(x - x_0) \dots (x - x_{i-1})(x - x_{i+1}) \dots (x - x_n)}{(x_i - x_0) \dots (x_i - x_{i-1})(x_i - x_{i+1}) \dots (x_i - x_n)} \quad (8.8)$$

This formula is called the Lagrange interpolation polynomial.

From formula (8.8) we can obtain expressions for linear ($n=1$) and quadratic ($n=2$) interpolations:

$$L(x) = \frac{x - x_1}{x_0 - x_1} y_0 + \frac{x - x_0}{x_1 - x_0} y_1, \quad n = 1$$

$$L(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \\ + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2 \quad n = 2$$